Nondimensional Staged Feedback Control of Mitochondrial Heteroplasmy Dynamics with Host-Emulated Verification on an Ultra-Low-Power `RV32IMF_Zve32x` Neural Processing Unit

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Abstract

Restoring bioenergetic function in cells burdened by mutant or damaged mitochondrial populations (D) alongside wild-type populations (W) requires navigating nonlinear couplings among autophagic clearance bottlenecks (B), reactive oxygen species (ROS) conversion, and replicative selection ($k_{\text{rep,adv}} \geq 1$). Here we formulate an eight-state ordinary differential equation (ODE) model with exact 1-to-1 correspondence between analytical equations, IEEE-754 `FP64` simulation, and bare-metal C99 firmware, analyzing a three-input staged feedback policy: Stage 0 bottleneck clearance (u_0), Stage 1 selective mitophagy (u_1), and Stage 2 biogenesis (u_2) gated jointly at heteroplasmy $h(t) = D(t)/[W(t) + D(t)] \leq h_{\text{gate}} = 0.080$ and bottleneck load $B(t) \leq B_{\text{gate}} = 0.120$. Using Nagumo's subtangent condition, we prove joint forward invariance of the Stage-1 safe set (EP $\leq 1.0456 < \theta_{\text{GSH}} = 1.150$, $W_{\min} = 0.600 > \theta_{\text{surv}} = 0.450$, contraction floor $\lambda_{\min} = 0.01085\ \text{d}^{-1}$) and derive both an exact accumulated-selection identity $s(t) = \int_0^t \lambda_h(q)\,dq$ (supremum error 0.0000, $s^*(0.40, 0.08) = 2.0369$) and a zero-tuning closed-form perturbation bound for constant-rate rescaling $\hat{\tau} = \hat{\lambda}_0 t$ across three non-homothetic regimes ($R^2 = 0.9828$, supremum error $0.0493 \leq 0.1216$). Solving the 300-variable discrete-adjoint optimal control problem (`L-BFGS-B`, gradient error 1.95×10^{-9}, 5-start spread 0.12%) shows that Pontryagin-optimal trajectories ride an interior mitophagy arc $u_1^*(t) \in (0, 1)$ along EP $= \theta_{\text{GSH}}$ before a terminal biogenesis sprint ($J^*_{\text{PMP}} = 0.7262$ vs. two-stage relay $J^*_{\text{2stage}} = 0.850$ and threshold feedback $J_{\text{thresh}} = 0.8875$), trading +22.21% open-loop Bolza cost for a 20.1% redox safety margin. Mapping the 31×31 (W_0, k_{dmg}) phase plane delineates irreversible collapse (18.9%) and reconciles build-first rescue (2.9%) with purge-then-build control. Under four-channel proxy sensing ($\kappa(O_7) = 745.6$), ±20% 7-rate mismatch, ±8% initial-state uncertainty, AR(1) noise, and 5% 6.5σ spikes, margin-tightened Huber-EKF ($\hat{h}_{\text{gate}} = 0.065$) reduces premature switches from 24.17% to 5.00%. On host-emulated Google Coral NPU (`RV32IMF_Zve32x`, static workspace 3,696 B/8,192 B in L1 DTCM Bank 0), our C99 fixed-point Huber-EKF (0.94% $\hat{h}$ RMSE) and `Q16.16` `RK4` kernel with `Q8.24` slopes and $\Sigma\Delta$ remainder diffusion eliminate deadband stalling across a 16× step-size sweep (0.0534%–0.0563% vs. 100% naive `Q16.16` stall at $N = 8{,}000$).

1 Introduction

1.1 Nonlinear Coupling in Mitochondrial Quality Control

Eukaryotic bioenergetics depends on maintaining a high functional fraction of intact wild-type mitochondrial DNA and cristae (W) relative to deletion-bearing or depolarized mutant organelles (D) [1, 2, 3]. Across diverse biological timescales—including post-mitotic dopaminergic neurons in Parkinsonian neurodegeneration

($T_{\max}$ = 90 d) [4, 5], *ex vivo* chimeric antigen receptor (CAR) T-cell expansion bioreactors ($T_{\max}$ = 168 h) [6, 7, 8], and honeybee (*Apis mellifera*) thoracic flight muscle exposed to strobilurin Complex III inhibition ($T_{\max}$ = 14 d) [9, 10]—mitochondrial decline is governed by three interacting feedbacks:

1. **Upstream Clearance Bottlenecks (B):** Accumulation of α-synuclein oligomers, lipofuscin, or xenobiotic fungicides elevates energetic impedance and saturates autophagosome-lysosome competence $V_{\text{lys}}(B) = K_B/(K_B + B) \in (0, 1]$ [4, 11].
2. **Threshold ROS Conversion (EP $>$ θ_{GSH}):** When cellular energetic potential EP exceeds the glutathione/peroxiredoxin antioxidant buffering threshold θ_{GSH}, electron leak generates reactive oxygen species (ROS) that convert intact wild-type mass W into damaged mass D [12].
3. **Asymmetric Replicative Selection ($k_{\text{rep,adv}}$):** Deletion-bearing mtDNA templates or fragmented organelles frequently exhibit a replicative or survival advantage ($k_{\text{rep,adv}} \geq 1$) during non-selective biogenesis [2, 13].

1.2 Comparison with Prior Control Formulations

In a foundational mathematical study, Hoitzing et al. [14] analyzed stochastic and deterministic mtDNA population control in a two-species (W, D) system without an upstream lysosomal bottleneck state $B(t)$ or replicative advantage ($k_{\text{rep,adv}} = 1$). When wild-type mass W_0 is already near the minimum survival floor and $B_0 \approx 0$, Hoitzing et al. demonstrated that an initial biogenesis pulse ("build-first") prevents transient ATP starvation before selective clearance. Conversely, when an upstream bottleneck $B_0 > 0$ elevates bioenergetic impedance and $k_{\text{rep,adv}} > 1$, stimulating biogenesis ($u_2 > 0$) prior to clearing B and D drives the energy potential EP(t) above θ_{GSH} and expands the mutant pool. Defining the exact mathematical boundary between these regimes—and implementing a verifiable closed-loop controller within the strict memory and arithmetic constraints of an ultra-low-power neural processing unit (NPU)—remains an open systems-and-control problem.

1.3 Contributions and Manuscript Organization

This study is a strictly theoretical and host-emulated computational investigation. All complete 8-state differential equations, auxiliary algebraic rate definitions, and step-by-step mathematical proofs are provided in Appendix A so that the main text focuses clearly on the governing dynamics and control results:

1. **Core Organelle Balance and Joint Safety Invariance (§2.1–§2.2, Appendices A.1–A.2):** We formulate an 8-state bioenergetic plant unified 1-to-1 across LaTeX, Python (`eval_rhs_fp64`), and C99 (`coralnpu_eval_core8_rhs_q16`), prove joint forward invariance of the Stage-1 safe box via Nagumo's subtangent theorem (EP $\leq 1.0456 < \theta_{\text{GSH}}$, $W_{\min} = 0.600 > \theta_{\text{surv}} = 0.450$, Lemma 1), and establish regional exponential contraction ($\lambda_{\min} = 0.01085\ \text{d}^{-1}$, Proposition 1).
2. **Discrete-Adjoint Optimal Control and 2D Controllability Phase Plane (§2.3–§2.4):** Solving a 300-variable discrete-adjoint optimal control problem across 5 initializations (0.12% cost spread) decomposes the optimality gap among Pontryagin interior-arc control ($J^*_{\text{PMP}} = 0.7262$), optimal two-stage constant relay ($J^*_{\text{2stage}} = 0.850$), and closed-loop threshold feedback ($J_{\text{thresh}} = 0.8875$), while a 31×31 (W_0, k_{dmg}) phase diagram maps irreversible collapse (18.9%) vs. build-first rescue (2.9%).
3. **Accumulated-Selection Reduction Across Timescales (§3, Appendix A.3):** We separate exact accumulated-selection rescaling $s(t) = \int_0^t \lambda_h(q)\, dq$ (0.0000 error) from zero-tuning constant-rate rescaling $\hat{\tau} = \hat{\lambda}_0 t$ across three non-homothetic regimes, verifying the analytical error bound in every regime (Proposition 2, Table 2).

4. **Mismatched Huber-EKF Observability and `RV32IMF_Zve32x` Firmware (§4–§5, Appendix A.4):** We audit pre-switch 7-state Observability Gramian conditioning ($\kappa(O_7) = 745.6$), benchmark Python and C99 fixed-point Huber-EKF under ±20% kinetic mismatch and 6.5σ spikes, prove a reversible Markov persistence bound (Proposition 3) with continuous-plant actuator rollback ($\Delta h_{\max} = +0.0098$), and verify single-bank L1 DTCM residency (3,696 B) and deadband-free $\Sigma\Delta$ `RK4` integration (0.0534%–0.0563% error).

2 Core Population Dynamics and Staged Control Analysis

2.1 Organelle Population Balance and Indirect Observability

We model the intracellular mitochondrial network over $t \in [0, T_{\max}]$ via an eight-dimensional state vector $\mathbf{x}(t) = [W,\, D,\, B,\, E,\, G,\, N,\, C,\, M_8]^\top \in \mathbb{R}^8_{\geq 0}$, actuated by bounded controls $\mathbf{u}(t) = [u_0(t), u_1(t), u_2(t)]^\top \in [0,1]^3$ representing upstream bottleneck clearance (u_0), PINK1/Parkin selective mitophagy (u_1), and PGC-1α/TFAM mitochondrial biogenesis (u_2). Let $M(t) = W(t) + D(t)$ denote total organelle mass and $h(t) = D(t)/M(t) \in [0, 1)$ denote the mutant heteroplasmy fraction.

While all eight coupled differential equations and closed-form auxiliary rates (ATP, impedance $\dot{e}_R$, effective load f_{eff}, energy potential EP, and cofactor states E, G, N, C, M_8) are specified in Appendix A.1 and parameterized in Table 1, the core population selection dynamics are governed by the (W, D, B) subsystem:

$$\dot{W} = k_{\text{bio}}(u_2, N)\, W\left(1 - \frac{M}{K}\right) - \Big(k_{\text{mito},W} + k_{\text{dmg}} \max\big(0,\, \text{EP}(\mathbf{x}, \mathbf{u}) - \theta_{\text{GSH}}\big)\Big) W, \tag{1}$$

$$\dot{D} = k_{\text{rep,adv}}\, k_{\text{bio}}(u_2, N)\, D\left(1 - \frac{M}{K}\right) - k_{\text{mito},D}(u_1)\, V_{\text{lys}}(B)\, D + k_{\text{dmg}} \max\big(0,\, \text{EP}(\mathbf{x}, \mathbf{u}) - \theta_{\text{GSH}}\big) W, \tag{2}$$

$$\dot{B} = b_{\text{basal}} + k_{\text{acc}}\, D\, (1 - u_0) - k_{\text{clr}}\, u_0\, B. \tag{3}$$

Here $K > 1$ is the carrying capacity, $k_{\text{bio},0} = k_{\text{mito},W}/(1 - 1/K)$ calibrates the unstimulated homeostatic equilibrium to $(W^*, D^*) = (1.00, 0)$ ($k_{\text{bio},0} = 0.075\ \text{d}^{-1}$ for $K = 1.250$, $k_{\text{mito},W} = 0.015\ \text{d}^{-1}$), and $\text{EP}(\mathbf{x}, \mathbf{u}) = \dot{e}_R(\mathbf{x})(1 - c_1 u_1 + c_2 u_2)^2$ couples proteostatic bottleneck load B and heteroplasmy h to supra-threshold ROS damage above θ_{GSH}.

Because (W, D, B) are not directly measurable at embedded sampling rates, the controller observes a discrete-time four-channel proxy $\mathbf{y}_k = [E(t_k),\, \text{ATP}(\mathbf{x}(t_k)),\, G(t_k),\, N(t_k)]^\top + \mathbf{v}_k$ (Eq. A.9 in Appendix A.1). At any single time instant t_k, the product $\text{EP} \cdot \text{ATP}/f_{\text{eff}}^2 = 1 + \beta_B B + \beta_h h$ depends on (B, h) only through a single linear combination, while (W, D) and phosphagen buffer C exhibit multiplicative iso-ATP trade-offs. Reconstructing $(\hat{W}, \hat{D}, \hat{B})$ therefore requires dynamic multi-step state estimation via a 7-state Huber-regularized extended Kalman filter (Huber-EKF) [15, 16], evaluated under ±20% plant-model mismatch in §5.2.

2.2 Heteroplasmy Quotient Flow and Stage-1 Safety Invariance

Differentiating $h(t) = D(t)/[W(t) + D(t)]$ along Eqs. (1)–(2) (full quotient-rule algebra in Appendix A.2) yields the exact heteroplasmy evolution law:

$$\dot{h}(t) = -\lambda_h(t)\, h(t)\big(1 - h(t)\big) + \big(1 - h(t)\big)\, k_{\text{dmg}} \max\big(0,\, \text{EP}(\mathbf{x}, \mathbf{u}) - \theta_{\text{GSH}}\big), \tag{4}$$

$$\lambda_h(t) = k_{\text{mito},D}(u_1)\, V_{\text{lys}}(B) - k_{\text{mito},W} - (k_{\text{rep,adv}} - 1)\, k_{\text{bio}}(u_2, N)\left(1 - \frac{M(t)}{K}\right). \tag{5}$$

Table 1: Canonical Neuronal Model Parameters, Physical Units, and Provenance Classification ($T_{\max} = 90$ d, matching `DEFAULT_NEURON_PARAMS` in Python and `p_q16` in C99). Tags: **M** (measured literature scale), **F** (fitted to equilibrium/turnover constraints), **A** (control design specification).

Symbol	Physical Description	Value	Units	Tag	Literature Basis / Calibration
K	Mitochondrial carrying capacity	1.250	Norm. mass	**F**	25% reserve above $W^* = 1.00$ [2]
$k_{\text{mito},W}$	Basal wild-type mitophagy rate	0.015	d^{-1}	**M**	Neuronal mtDNA $t_{1/2} \approx 46$ d [3, 4]
$k_{\text{bio},0}, k_{\text{bio,max}}$	Basal / max stimulated biogenesis	0.075, 0.140	d^{-1}	**F**	Basal $k_{\text{mito},W}/(1 - 1/K)$ [11]
$k_{\text{mito},D,\max}$	Max Stage-1 selective mitophagy	0.115	d^{-1}	**M**	PINK1/Parkin clearance ($t_{1/2} \approx 6$ d) [11]
$k_{\text{rep,adv}}$	Mutant replicative advantage	1.320	Dimensionless	**M**	Deletion mtDNA bias (swept 1.00–1.32) [2, 13]
k_{dmg}	Supra-threshold ROS damage rate	0.040	$\text{d}^{-1}\text{EP}^{-1}$	**F**	Swept 0.00–0.045 in Fig. 2B [4]
$\theta_{\text{GSH}}, \theta_{\text{cys}}$	Redox & thiol oxidation ceilings	1.150, 0.850	Norm. EP	**F**	Glutathione/peroxiredoxin ceiling [5]
θ_{surv}	Bioenergetic ATP survival floor	0.450	Norm. ATP	**M**	Bioenergetic failure floor (45%) [5]
K_B	Lysosomal bottleneck half-sat.	0.250	Norm. load	**F**	Autophagic flux saturation [4]
$b_{\text{basal}}, k_{\text{acc}}, k_{\text{clr}}$	Bottleneck influx, coupling, clear.	0.0072, 0.080, 0.180	d^{-1}	**F**	$b_{\text{basal}} = 472/2^{16}$; $B_{\text{eq}} = 0.040$
$\beta_B, \beta_h, \eta_D, \gamma_C$	Impedance & OXPHOS couplings	0.25, 0.65, 0.15, 0.34	Dimensionless	**F**	Defective OXPHOS coupling of D [1]
c_1, c_2	Mitophagy relief / biogenesis load	0.220, 0.040	Dimensionless	**F**	Energetic load modulation
$k_E, k_{G,\text{prod}}, k_{G,\text{clr}}$	Potential & stress readout rates	0.250, 0.080, 0.160	d^{-1}	**F**	First-order biomarker kinetics
$k_{N,\text{syn},0/2}, k_{\text{PARP}}, k_{N,\text{use}}$	NAD^+ synthesis, PARP1, use	0.060, 0.080, 0.060, 0.095	d^{-1}	**F**	PARP1 drain above θ_{GSH} [11]
$k_{\text{oct},0/2}, k_{\text{ox}}, k_{\text{drift}}, k_{\text{TET}}$	Buffer (C) & epigenetic (M_8)	0.04, 0.10, 0.05, 0.0015, 0.012	d^{-1}	**F**	Thiol oxidation & demethylation
$h_{\text{gate}}, B_{\text{gate}}$	Stage-1 $\rightarrow$ 2 switching gates	0.080, 0.120	Dimensionless	**A**	Tightened to $\hat{h}_{\text{gate}} = 0.065$ under EKF (§5.2)

Equations (4)–(5) expose the core dynamical conflict: when $k_{\text{rep,adv}} > 1$ and $B(t)$ is elevated, activating biogenesis prematurely ($u_2 = 1$) subtracts $(k_{\text{rep,adv}} - 1)k_{\text{bio}}(1 - M/K)$ from the selective clearance rate $\lambda_h(t)$ while simultaneously raising f_{eff} and pushing EP above θ_{GSH}, igniting positive ROS conversion of W into D. Conversely, Stage 1 clearance ($u_0 = 1, u_1 = 1, u_2 = 0$) unblocks $B(t)$ and lowers f_{eff}, preserving joint forward invariance of a safe physiological box:

Lemma 1 (Joint Stage-1 Safe-Box Invariance; Proof in Appendix A.2)**.** *Under Stage 1 control ($u_0 = 1, u_1 = 1, u_2 = 0$) on the compact candidate box $\Omega_1(D_0, B_0) = \{\mathbf{x} \in \mathbb{R}^8_{\geq 0} : W \in [W^* - D_0, K],\ D \in [0, D_0],\ B \in [B_{\text{eq}}, B_0],\ E \in [0, E_{\max}],\ C \in [C_{\min}, 1.30]\}$ with $\sup_{\Omega_1} \text{EP}(\mathbf{x}, (1, 1, 0)) \leq \min(\theta_{\text{GSH}}, E_{\max})$ and critical bottleneck threshold B^*_{crit} defined in Eq. (A.11):*

1. ***Joint Forward Invariance** ($B_0 \leq B^*_{\text{crit}}$): $\Omega_1(D_0, B_0)$ is positively invariant; throughout $t \in [0, t_{\text{gate}}]$, ROS conversion remains identically zero ($\text{EP} \leq \theta_{\text{GSH}}$), $D(t) \leq D_0$, and wild-type mass never drops below $W_{\min}(D_0) = W^* - D_0$. On the canonical neuronal box ($D_0 = 0.400, B_0 = 0.500 < B^*_{\text{crit}} = 1.2020$), $\max_{\Omega_1} \text{EP} = 1.0456 < \theta_{\text{GSH}} = 1.150$ and $W(t) \geq 0.600 > \theta_{\text{surv}} = 0.450$.*
2. ***Super-Critical Dwell Time** ($B_0 > B^*_{\text{crit}}$): Under $u_0 = 1$, $B(t)$ decays exponentially to B^*_{crit} in finite time $t_0 = k_{\text{clr}}^{-1} \ln[(B_0 - B_{\text{eq}})/(B^*_{\text{crit}} - B_{\text{eq}})]$, after which Case 1 invariance holds whenever $\bar{D}(t_0) < W^*$ ($t_0 = 2.91$ d at $B_0 = 2.00$).*

Proposition 1 (Regional Contraction and Invasion Stability; Proof in Appendix A.2)**.** *For any initial state $\mathbf{x}(0) \in \Omega_1(D_0, B_0)$ satisfying Lemma 1 Case 1, the selective contraction rate $\lambda_h(t)$ is bounded below uniformly on $[0, t_{\text{gate}}]$ by a closed-form constant $\lambda_{\min}$ (Eq. A.12; $\lambda_{\min} = 0.01085\ d^{-1} > 0$ on the canonical neuronal box). Whenever $\lambda_{\min} > 0$, heteroplasmy contracts monotonically according to*

$$h(t) \leq \frac{h_0\, e^{-\lambda_{\min} t}}{1 - h_0 + h_0\, e^{-\lambda_{\min} t}} \leq h_0 \exp\big(-\lambda_{\min}(1 - h_0)t\big). \tag{6}$$

Moreover, at the mutant-free homeostatic equilibrium $(W^*, 0, B_{\text{eq}})$, the linearized mutant invasion eigenvalue $\mu_D = k_{\text{rep,adv}} k_{\text{mito},W} - k_{\text{mito},D}(u_1) V_{\text{lys}}(B_{\text{eq}})$ switches sign from unstable ($\mu_D > 0$ when unstimulated) to locally exponentially stable ($\mu_D < 0$) under Stage-1 mitophagy.

2.3 Discrete-Adjoint Optimal Control vs. Embedded Threshold Feedback

To benchmark simple threshold switching against an unconstrained time-varying control schedule identically to `compute_bolza_cost` in `simulator/generate_paper1_verified_benchmarks.py`, we minimize the finite-horizon Bolza cost functional:

$$J(\mathbf{u}) = w_h\, h(T_{\max})^2 + w_W \big(W_{\text{ref}} - W(T_{\max})\big)^2 + \int_0^{T_{\max}} \Big[w_{\text{EP}} \max\big(0, \text{EP}(t) - \theta_{\text{GSH}}\big)^2 + w_{\text{surv}} \max\big(0, \theta_{\text{surv}} - \text{ATP}(t)\big)^2 + \frac{w_{\text{def}}}{T_{\max}} \max\big(0, 0.85 - W(t)\big)^2 + r_1 u_1(t) + r_2 u_2(t) \Big]\, dt, \tag{7}$$

with terminal weights $w_h = 80$, $w_W = 30$ ($W_{\text{ref}} = 1.00$), safety weights $w_{\text{EP}} = 180\ \text{d}^{-1}$, $w_{\text{surv}} = 250\ \text{d}^{-1}$, $w_{\text{def}} = 0.85$, and horizon-normalized actuation weights $r_1 = 1.15/T_{\max}$, $r_2 = 0.12/T_{\max}$ ($r_0 = 0 \implies u_0^*(t) \equiv 1$).

Discretizing $\mathbf{u}(t) \in [0,1]^3$ into $N_{\text{ctrl}} = 100$ piecewise-constant intervals (300 decision variables over 1,000 exact `RK4` steps on $[0, 90\ \text{d}]$) and propagating exact discrete-time `RK4` adjoint Jacobians (central finite-difference relative error 1.95×10^{-9}), `L-BFGS-B` converges across 5 diverse initializations within a 0.12% cost spread ($J^*_{\text{PMP}} = 0.7262$, $h(T_{\max}) = 0.0300$, $W(T_{\max}) = 0.9938$; Figure 2A). Rather than pure bang-bang switching, $u_1^*(t)$ modulates smoothly along an early interior redox arc ($[0.44, 0.83]$ on Days 0–17) so $\text{EP}^*(t)$ rides $\theta_{\text{GSH}} = 1.150$, while $u_2^*(t)$ remains zero until $h^*(t) < 0.030$ before executing a terminal biogenesis sprint (Days 78–90).

Because riding $\text{EP}^*(t) \equiv \theta_{\text{GSH}}$ in open loop leaves zero robustness margin against kinetic mismatch, our embedded closed-loop law triggers Stage 1 $\to$ Stage 2 ($(1,1,0) \to (1, 0.35, 1)$) when $B(t) \le B_{\text{gate}} = 0.120$ and $h(t) \le h_{\text{gate}} = 0.080$ (Eq. A.10 in Appendix A.1). At $h_{\text{gate}} = 0.080$ ($t_{\text{gate}} = 33.30$ d), post-switch $\text{EP}(t^+_{\text{gate}})$ steps from 0.603 to 0.919 < 1.150, preserving a 20.1% redox margin while incurring a decomposed +17.05% relay-restriction cost ($J^*_{\text{2stage}} = 0.850$) plus a +4.38% constant-threshold cost ($J_{\text{thresh}} = 0.8875$; +22.21% total vs. J^*_{PMP}, Figure 1B).

2.4 Competing Policies, Hoitzing Reconciliation, and 2D Controllability Phase Plane

Table 3A and Figure 1A compare six control strategies from the canonical neuronal state ($W_0 = 0.60, D_0 = 0.40, B_0 = 0.50$): Untreated ($W(T_{\max}) = 0$), Concurrent ($\text{EP}_{\max} = 1.423 > \theta_{\text{GSH}}$, $J = 99.83$), Build-First at $B_0 = 0.50$ ($W(T_{\max}) = 0$), Staged Threshold ($W(T_{\max}) = 1.0487$, $h(T_{\max}) = 0.0249$, $J = 0.8875$), Two-Stage Relay Optimum ($J = 0.850$), and Adjoint Optimum ($J = 0.7262$).

To reconcile this result with Hoitzing et al. [14]—who studied low-wild-type populations with partial mutant OXPHOS ($\eta_D = 0.65$), $k_{\text{rep,adv}} = 1.00$, and negligible ROS/bottleneck load ($k_{\text{dmg}} = 0, B_0 = 0.02$)—we integrated both Build-First and Purge-Then-Build across a 31×31 grid (961 points, 1,922 runs) spanning $W_0 \in [0.12, 0.42]$ and $k_{\text{dmg}} \in [0.000, 0.045]\ \text{d}^{-1}\text{EP}^{-1}$ ($D_0 = 0.38, B_0 = 0.15, \eta_D = 0.65, k_{\text{rep,adv}} = 1.15$). Figure 2B resolves four topological zones meeting near $(W_0, k_{\text{dmg}}) \approx (0.185, 0.023)$:

1. **Zone 0 — Irreversible Bioenergetic Collapse** ($182/961 = 18.9\%$)**:** Severe initial depletion violates $\text{ATP}(t) \ge \theta_{\text{surv}} = 0.450$ or prevents recovery under *both* policies.
2. **Zone 1 — Hoitzing Build-First Exclusive Rescue** ($28/961 = 2.9\%$)**:** A low-W_0, weak-ROS wedge ($W_0 \in [0.15, 0.19], k_{\text{dmg}} \le 0.023$) where Build-First holds $\text{ATP}(t) \ge 0.450$ while Purge-First dips transiently to $\text{ATP}_{\min} = 0.440$.
3. **Zone 2 — Purge-Then-Build Exclusive Rescue** ($59/961 = 6.1\%$)**:** At $k_{\text{dmg}} \ge 0.023$, Build-First drives $\text{EP}(t) > \theta_{\text{GSH}}$ and triggers runaway wild-type loss, whereas Purge-Then-Build rescues the cell.

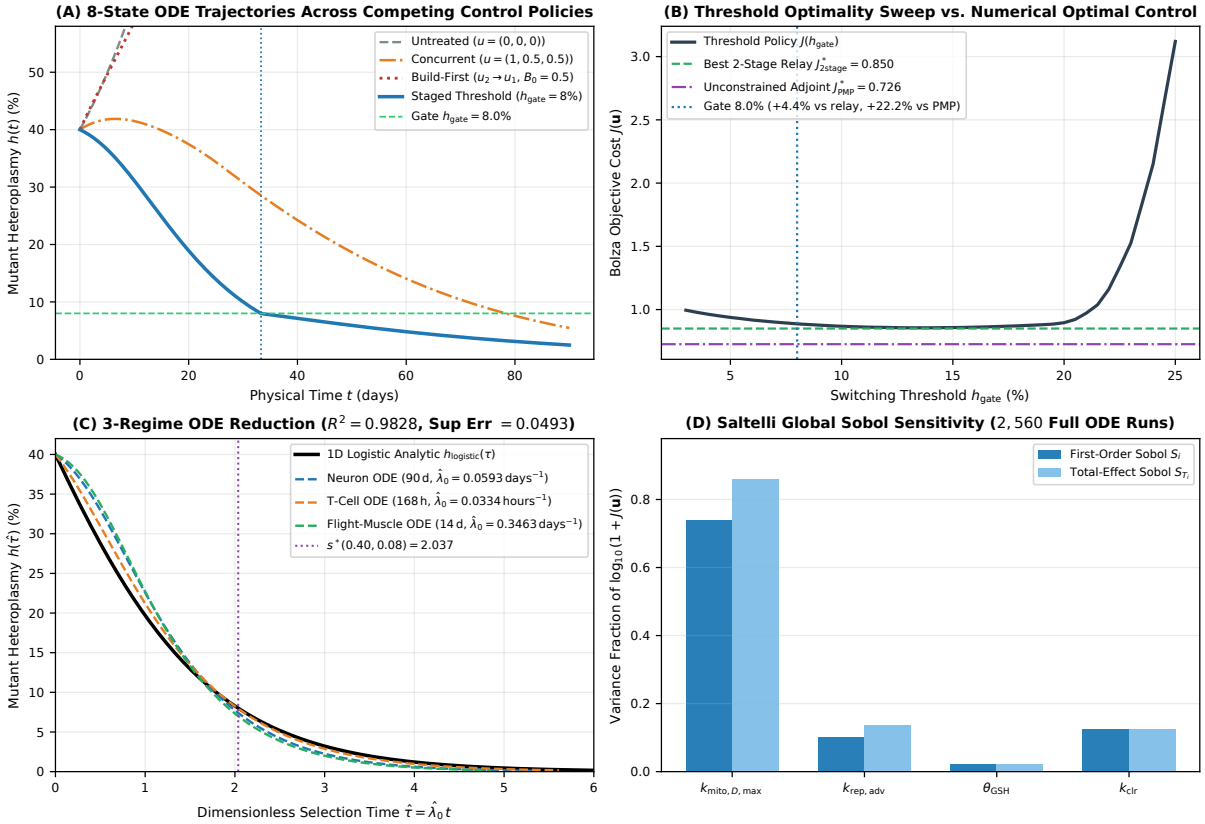


Figure 1: **First-Principles 8-State ODE Control Comparison, Threshold Optimality Sweep, Independent 3-Regime Reduction, and Saltelli Sobol Sensitivity Analysis** (seed 20261109). **(A)** Mutant heteroplasmy trajectories $h(t)$ under Untreated, Concurrent, Build-First ($B_0 = 0.50$), and Staged Threshold ($h_{\text{gate}} = 8.0\%, B_{\text{gate}} = 0.120$, switching at $t_{\text{gate}} = 33.30$ d) policies. **(B)** Direct sweep of $J(h_{\text{gate}})$ across $h_{\text{gate}} \in [3\%, 25\%]$ compared against the optimal 2-stage constant relay ($J^*_{\text{2stage}} = 0.850$, +4.38%) and unconstrained 300-variable discrete-adjoint optimal control ($J^*_{\text{PMP}} = 0.7262$, +22.21%). **(C)** Independent 8-state ODE trajectories for Neuronal (90 d), T-Cell (168 h), and Flight-Muscle (14 d) regimes plotted against closed-form constant-rate dimensionless time $\hat{\tau} = \hat{\lambda}_0 t$ ($R^2 = 0.9828$, supremum error 0.0493) and exact accumulated selection $s^* = 2.0369$. **(D)** Global variance-based Sobol sensitivity indices (S_i, S_{T_i}) across 2,560 ODE integrations.

4. **Zone 3 — Both Safe, Purge-Then-Build Optimal** (692/961 = 72.0%)**:** Both policies maintain $\text{ATP}(t) \geq 0.450$, with Purge-Then-Build achieving strictly lower Bolza cost.

3 Accumulated-Selection Reduction Across Biological Timescales

Dimensional analysis alone [17] cannot collapse an 8-state nonlinear ODE onto a 1D curve unless the time-variation of $\lambda_h(t)$ in Eq. (5) is explicitly accounted for. We therefore establish both the exact accumulated-selection identity and a zero-tuning closed-form perturbation bound for constant-rate time rescaling.

Proposition 2 (Accumulated-Selection Reduction; Proof & Closed-Form Integral in Appendix A.3)**.** *Suppose* $\text{EP}(t) \leq \theta_{\text{GSH}}$ *on* $[0, t]$ *during Stage 1* ($u_0 = 1, u_1 = 1, u_2 = 0$) *with* $h_0 \in (0, 1)$.

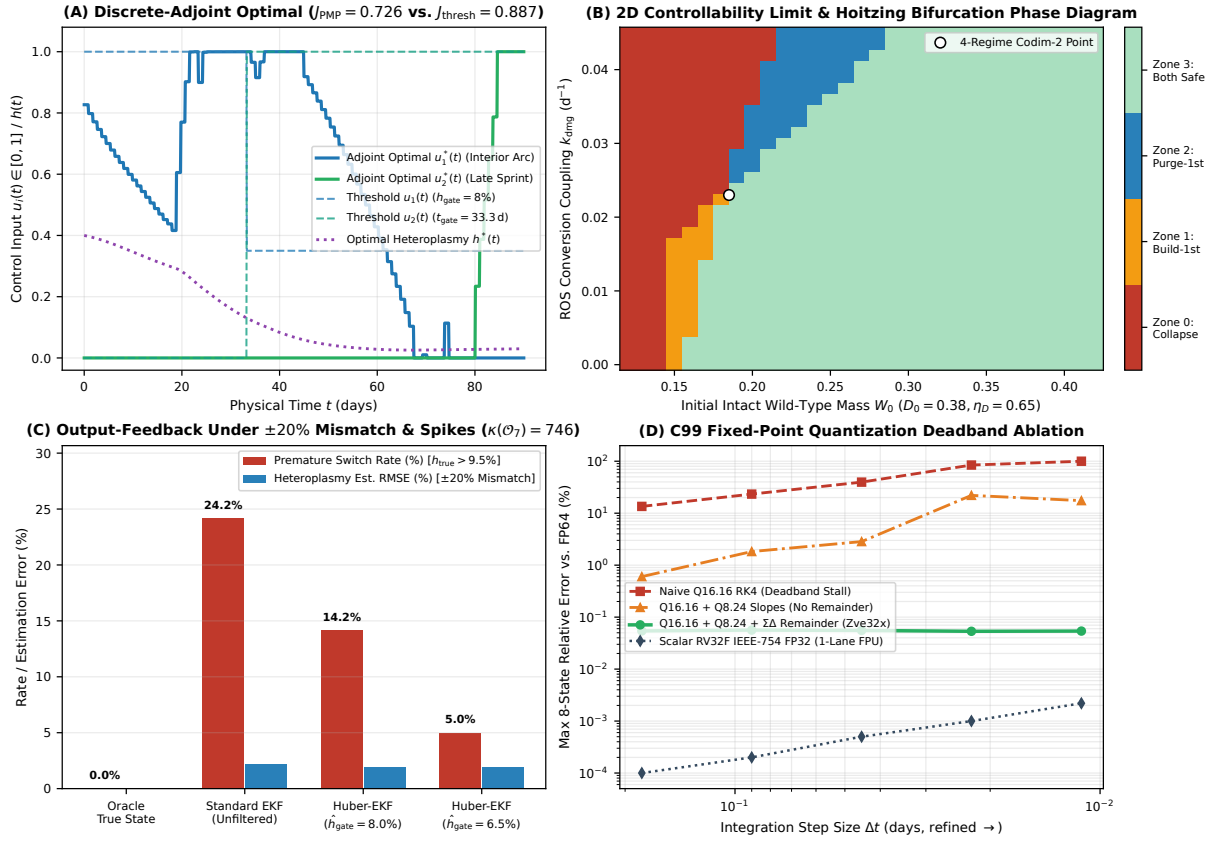


Figure 2: **Unconstrained Discrete-Adjoint Optimal Control, 2D Controllability Phase Diagram, Mismatched Closed-Loop Huber-EKF Audit, and C99 Fixed-Point Deadband Ablation.** **(A)** 300-variable `L-BFGS-B` discrete-adjoint optimal controls $(u_1^*(t), u_2^*(t))$ ($J^*_{\mathrm{PMP}} = 0.7262$, 5-start spread 0.12%) vs. closed-loop staged threshold control ($J_{\mathrm{thresh}} = 0.8875$). **(B)** 31×31 phase diagram over (W_0, k_{dmg}) mapping Zone 0 Irreversible Collapse (18.9%), Zone 1 Build-First Exclusive Rescue (2.9%), Zone 2 Purge-First Exclusive Rescue (6.1%), and Zone 3 Both Safe (72.0%). **(C)** 120-trial closed-loop Output-Feedback Monte Carlo benchmark under ±20% 7-rate mismatch, ±8% initial-state uncertainty, AR(1) noise ($\rho = 0.60$), and 5% 6.5σ spikes ($\kappa(\mathcal{O}_7) = 745.6$). **(D)** Multi-step-size C99 quantization deadband ablation ($\Delta t \in [0.18, 0.01125]$ d) comparing naive `Q16.16 RK4`, `Q8.24` slopes without remainder, `Q8.24` slopes with 8-lane ΣΔ remainder diffusion, and scalar `RV32F FP32`.

1. ***Exact Accumulated-Selection Coordinate:*** *In terms of the dimensionless selection integral* $s(t) \triangleq \int_0^t \lambda_h(q)\, dq$*, the 8-state heteroplasmy trajectory satisfies the exact logistic law*

$$h(t) = h_{\mathrm{logistic}}\big(s(t); h_0\big) \triangleq \frac{h_0\, e^{-s(t)}}{1 - h_0 + h_0\, e^{-s(t)}}, \qquad s^*(h_0, h_{\mathrm{gate}}) = \ln\!\left(\frac{h_0(1 - h_{\mathrm{gate}})}{h_{\mathrm{gate}}(1 - h_0)}\right), \tag{8}$$

yielding $s^*(0.40, 0.08) = 2.0369$*,* $s^*(0.25, 0.08) = 1.3437$*, and* $s^*(0.60, 0.08) = 2.8478$*.*

2. ***Zero-Tuning Perturbation Bound for Constant-Rate Rescaling*** $\hat{\tau} = \hat{\lambda}_0 t$***:*** *Let* $\hat{\lambda}_0$ *denote the closed-form a priori Stage-1 contraction rate obtained by integrating* $V_{\mathrm{lys}}(B(q))$ *analytically over the cleared-bottleneck horizon estimate (Eqs. A.13–A.14 in Appendix A.3, with zero fitted constants). Then for all*

Table 2: Three-Regime 8-State ODE Verification of Proposition 2 Across Distinct Dimensionless Π-Groups ($\Pi_1 = k_{\text{mito},W}/k_{\text{mito},D,\max}, k_{\text{rep,adv}}, K, K_B, B_0, \eta_D$). Each regime uses the zero-tuning closed-form rate $\hat{\lambda}_0$ (Appendix A.3) and satisfies Eq. (9).

ODE Regime ($T_{\max}$)	**Distinct Π-Groups**	$\hat{\lambda}_0$	t_{gate} ($h_0 = 0.4$)	$\hat{\tau}_{\text{gate}}$ (0.40/0.25/0.60*)	**Exact** $s(t)$ **Err**	**Sup Err vs. Bound**	**RMSE**	R^2
1. Neuron (90.0 d)	(0.130, 1.32, 1.25, 0.25, 0.50, 0.15)	0.05928 d^{-1}	33.30 d	1.974/1.414/5.336	0.0000	0.0422 ≤ 0.1216	0.0168	0.9812
2. CAR-T (168.0 h)	(0.125, 1.12, 1.40, 0.35, 0.28, 0.28)	0.03340 h^{-1}	60.82 h	2.031/1.403/3.024	0.0000	0.0216 ≤ 0.0399	0.0084	0.9946
3. *Apis* (14.0 d)	(0.125, 1.22, 1.18, 0.20, 0.60, 0.08)	0.34632 d^{-1}	5.57 d	1.930/1.396/4.848	0.0000	0.0493 ≤ 0.1213	0.0203	0.9745
1D Analytic Logistic	EP(t) ≤ θ_{GSH} ($s^* = \hat{\tau}_{\text{gate}}$)	1.00000	2.0369	2.037/1.344/2.848	—	**All 3 Hold (Max** 0.0493)	**0.0160**	**0.9828**

$t \in [0, t_{\text{gate}}]$, the constant-rate approximation error satisfies

$$\left|h(t) - h_{\text{logistic}}(\hat{\lambda}_0 t; h_0)\right| \le \frac{1}{4}\Big(\mathcal{E}_B(t) + \mathcal{E}_M(t)\Big), \tag{9}$$

where $\mathcal{E}_B(t)$ and $\mathcal{E}_M(t)$ bound transient bottleneck-clearance and total-mass variations (Eq. A.15).

Table 2 and Figure 1C confirm Proposition 2 across three genuinely non-homothetic parameter sets: at $h_0 = 0.40$, exact $s(t)$ achieves 0.0000 error and constant-rate $\hat{\tau} = \hat{\lambda}_0 t$ yields pooled $R^2 = 0.9828$ with supremum error strictly inside Eq. (9) in all three regimes. Conversely, at severe initial burden $h_0 = 0.60^*$ ($W_0 = 0.40, D_0 = 0.60$), EP(0) breaches $\theta_{\text{GSH}} = 1.150$ ($\text{EP}_{\text{peak}} = 1.312$–$9.734$), activating ROS damage in Eq. (4) and delaying $\hat{\tau}_{\text{gate}}$ to 3.024–5.336 > $s^*(0.60, 0.08) = 2.848$.

4 Embedded Fixed-Point Implementation on Coral NPU

4.1 Target Architecture and Single-Bank L1 DTCM Residency

We target the open-source Google Coral NPU architecture (`RV32IMF_Zve32x`) [18, 19], pairing a 32-bit RISC-V scalar core (with scalar single-precision `FP32`) and an integer-only 128-bit SIMD vector unit (`Zve32x`, four 32-bit or eight 16-bit integer lanes) with 8 KB ITCM and 32 KB DTCM split into four 8,192-byte banks. Constraining the entire runtime state (INT8 gate/Goertzel LUTs, mixed `Q16.16`/`Q8.24` 7-state Huber-EKF covariance matrices, `RK4` slope registers, 64-frame ring buffers, and stack) to `coralnpu_dtcm_bank0_workspace_t` occupies 3,696 **B** (45.12%) **of a single** 8,192**-byte bank** (`Bank 0`), allowing `Banks 1–3` (24 KB) to remain power-gated during chronic monitoring (Figure 3D, Table 6).

4.2 Widening Vector `RK4`, ΣΔ Remainder Diffusion, and Integer Huber-EKF

In `build/coralnpu/mitorestorer_coralnpu_kernel.h`, exact integer firmware parity across multi-week horizons combines three mechanisms:

1. **0-ULP 64-Bit Widening Vector Updates:** Multiplying `Q16.16` operands or scaling `Q8.24` slopes by Δt_{Q16} uses 32-to-64-bit widening SIMD multiplication (`__riscv_vwmul_vx_i64m4`) and narrowing arithmetic shift (`__riscv_vnsra_wx_i32m2`), achieving **0-ULP discrepancy** against scalar 64-bit integer reference across 2,048 randomized lane operations.
2. **`Q8.24` Slopes with 8-Lane ΣΔ Remainder Diffusion:** Slow biological turnover ($k_{\text{mito},W} = 0.015\ d^{-1}$) at small step sizes $\Delta t \le 0.045$ d produces per-step increments smaller than 0.5 LSB_{16} (7.63×10^{-6}), stalling naive `Q16.16` `RK4` completely (Figure 2D). Evaluating derivatives in `Q8.24` ($2^{-24} \approx 5.96\times10^{-8}$) and propagating the exact 64-bit numerator remainder $\mathbf{r}_k \in (-6 \cdot 2^{24}, 6 \cdot 2^{24})^8$ across steps eliminates quantization deadband drift across a 16× step sweep.

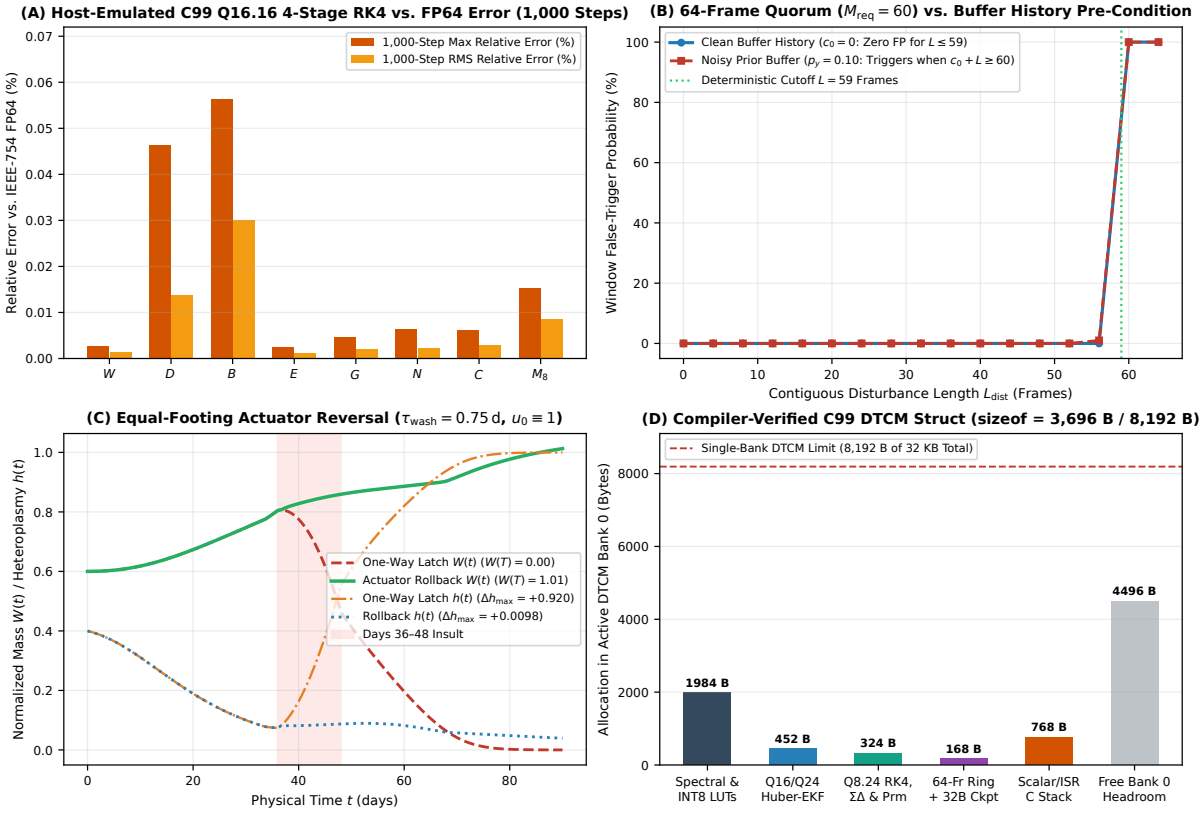


Figure 3: **Host-Emulated C99 `Q16.16` 4-Stage `RK4` Parity, 64-Frame Quorum Pre-Condition Analysis, Equal-Footing Plant-Preserving Rollback, and Verified L1 DTCM Footprint.** **(A)** Full-horizon (1,000-step) maximum and RMS relative error (%) between bare-metal C99 `Q16.16` `RK4` (`coralnpu_hil_testbench.c`) and IEEE-754 `FP64` across all 8 states (global max error 0.0563%). **(B)** 64-frame persistence false-trigger rate ($M_{\text{req}} = 60$) vs. contiguous disturbance length L_{dist} under clean ($c_0 = 0$: zero false triggers for $L_{\text{dist}} \leq 59$) vs. noisy ($p_y = 0.10$) prior buffers. **(C)** Equal-footing ($u_0 \equiv 1$) continuous-plant actuator reversal under an acute Days 36–48 exogenous surge ($\tau_{\text{wash}} = 0.75$ d), bounding $\Delta h_{\max} = +0.0098$ and rescuing $W(T_{\max}) = 1.0128$ vs. collapse ($W \to 0.0001$) under a one-way latch. **(D)** Compiler-verified static L1 DTCM Bank 0 workspace (3,696 B/8,192 B).

3. **Zero-Heap Mixed `Q16.16`/`Q8.24` Sequential Huber-EKF:** Without `malloc` or `<math.h>`, `coralnpu_huber_ekf7_meas_update_q16` stores $\hat{\mathbf{x}}_{0:6}$ in `Q16.16` and $\mathbf{P} \in \mathbb{Z}_{32}^{7\times 7}$ in `Q8.24`, processes the four proxy channels sequentially via scalar innovation variances $S_i = \mathbf{h}_i \mathbf{P} \mathbf{h}_i^{\top} + R_i$, computes $\sqrt{S_i}$ via exact 64-bit integer Newton square root (`coralnpu_isqrt64`), and applies symmetric Huber clipping $w_{\text{Hub}} = \min(1, \gamma_{\text{Huber}} \sqrt{S_i}/|\nu_i|)$.

5 Output-Feedback Observability, Disturbance Rejection, and Verification

5.1 64-Frame Dual-Modality Persistence Under Markov Noise

To prevent correlated transient spikes from triggering premature switching, a 64-frame bitmask $\mathcal{B}_k \in \{0,1\}^{64}$ requires $M_{\text{req}} = 60$ primary exceedances ($\mu_{\text{req}} = 60/64 = 0.9375$) plus a concurrent secondary modality check

Table 3: First-Principles Policy Comparison (1,000-Interval Bolza Integration, Including 5-Start 300-Variable Discrete-Adjoint Optimum) and 1,000-Stream Correlated AR(1) Persistence Benchmarks (seed `20261109`).

A. Neuronal 8-State Control Policy ($T_{\max}$ = 90 **d**)	**Bolza Cost** $J(\mathbf{u})$	**Switch** t_{gate}	**Term.** $h(T_{\max})$	**Term.** $W(T_{\max})$	**Peak** $\text{EP}_{\max}$	**Min** $\text{ATP}_{\min}$
Untreated Baseline ($\mathbf{u} = (0,0,0)$)	2,768,855.31	—	1.0000	0.0000	20.605	0.186 (Viol.)
Concurrent Biogenesis + Mitophagy ($\mathbf{u} = (1,0.5,0.5)$)	99.829	—	0.0546	0.9474	1.423	0.806 (Safe)
Build-First Policy in High-Impedance Regime ($B_0 = 0.50$)	760,388.40	—	1.0000	0.0000	9.744	0.028 (Viol.)
Proposed Staged Threshold ($h_{\text{gate}} = 0.080$, $B_{\text{gate}} = 0.120$)	**0.8875**	**33.30** d	**0.0249**	**1.0487**	**1.046**	**0.806 (Safe)**
Two-Stage Constant Relay Optimum ($t^*_{\text{sw}} = 48.18$ d)	0.8500 (J^*_{2stage})	48.18 d	0.0375	1.0421	1.046	0.806 (Safe)
Unconstrained 300-Var Discrete-Adjoint Optimum (`L-BFGS-B`)	0.7262 (J^*_{PMP})	Interior Arc	0.0300	0.9938	1.150	0.806 (Safe)
B. 1,000-Stream Correlated AR(1) **Noise Benchmark**	**Lag-1** ρ	**Burst** L_{dist}	**1-Sample FP**	**64-Quorum FP**	**Dual-Mod. FP**	**95% Upper Bd**
1. *Apis* Wind-Gust Disturbance ($\Delta t_{\text{frame}} = 1$ s)	0.50	12 frames	100.0%	0.00% (0/1000)	0.00%	$\leq 0.299\%$
2. CAR-T Lactate Biofouling Spike ($\Delta t_{\text{frame}} = 1$ min)	0.70	25 frames	100.0%	0.00% (0/1000)	0.00%	$\leq 0.299\%$
3. Wearable Postprandial Excursion ($\Delta t_{\text{frame}} = 15$ min)	0.60	45 frames	100.0%	1.10% (11/1000)	0.70%	$\leq 1.219\%$

$z_k \geq \gamma_{\text{sec}}$.

Proposition 3 (Markov Persistence Bound; Proof in Appendix A.4)**.** *Let stationary sensor noise evolve via the reversible Gaussian* AR(1) *Markov chain* $e_k = \rho e_{k-1} + \sqrt{1-\rho^2}\epsilon_k$ *($\rho \in [0,1)$, spectral gap $1-\rho$), with primary exceedance probability $p_y < \mu_{\text{req}} = 0.9375$, quorum-conditioned secondary probability $p_{z|Q}$, and $c_0 \in \{0,\ldots,64\}$ prior buffer exceedances before a contiguous burst of L_{dist} frames:*

1. ***Deterministic Sub-Quorum Cutoff:*** *False quorum is deterministically impossible whenever $L_{\text{dist}} \leq 59 - c_0$ ($L_{\text{dist}} \leq 59$ under a clean prior buffer $c_0 = 0$; Figure 3B).*
2. ***Spectral-Gap False-Trigger Bound:*** *By Paulin's concentration inequality for reversible Markov chains [20], single-step and K_{frames}-horizon false-trigger probabilities satisfy*

$$P_{\text{FP,step}} \leq p_{z|Q} \exp\left(-2N_{\text{win}} \frac{1-\rho}{1+\rho}(\mu_{\text{req}} - p_y)^2\right), \qquad P_{\text{FP,horizon}} \leq \min\left(1,\ K_{\text{frames}} P_{\text{FP,step}}\right). \tag{10}$$

In physical deployments, the 64-frame window $T_{\text{win}} = 64\,\Delta t_{\text{frame}}$ scales with plant kinetics: $\Delta t_{\text{frame}} = 1.0$ s for *Apis* wingbeat telemetry ($T_{\text{win}} = 64$ s), 1.0 min for CAR-T bioreactors (64 min), and 15.0 min for chronic neural monitoring (16.0 h), verified across 1,000 correlated AR(1) streams in Table 3B.

5.2 Stage-1 Observability Gramian, Mismatched Huber-EKF, and Continuous-Plant Rollback

Evaluating the empirical 7-state Observability Gramian O_7 strictly over the pre-switch Stage-1 window $t \in [0, 33.30\text{ d}]$ (370 steps, $\Delta t = 0.09$ d; $\sigma_8 = 0$ identically for downstream M_8) yields singular values 963.44, 253.33, 205.73, 171.19, 137.38, 15.61, and 1.29, with condition number $\kappa(O_7) = 745.6$, reflecting iso-ATP (C, W, D) compensation ($\sigma_7 = 1.29$) and (B, h) collinearity ($\sigma_6 = 15.61$). Across 120 closed-loop Monte Carlo trials with simultaneous ±20% 7-rate mismatch, ±8% initial-state uncertainty, AR(1) noise ($\rho = 0.60$), and 5% 6.5σ spikes (Figure 2C, Table 4A), unfiltered Standard EKF ($\hat{h}_{\text{gate}} = 0.080$) suffers 24.17% (29/120) premature switches ($h_{\text{true}} > 0.095$), Huber-EKF ($\gamma_{\text{Huber}} = 1.50$) + 8-step persistence cuts premature switches to 14.17% (17/120), and margin-tightened Huber-EKF ($\hat{h}_{\text{gate}} = 0.065$) reduces premature switches 4.8× to 5.00% (6/120, $\hat{h}$ **RMSE** 1.93%, **mean** $W(T_{\max}) = 1.0479$).

Crucially, restoring a microcontroller's internal checkpoint $\mathbf{x}_{\text{ckpt}}$ cannot reset the physical biological state $\mathbf{x}(t)$. Under an equal-footing ($u_0 \equiv 1$) Days 36–48 uncoupling/proteostatic surge ($\Delta b_{\text{basal}} = +0.52\ \text{d}^{-1}$, $c_2 = 0.32$, washout $\tau_{\text{wash}} = 0.75$ d, detection latency $\delta t_{\text{detect}} = 0.54$ d; Figure 3C), keeping Stage-2 biogenesis latched on drives $\text{EP}_{\text{peak}} = 11.128$ and collapses $W(T_{\max}) \to 0.0001$, whereas continuous-plant actuator

Table 4: **(A)** 120-Trial Closed-Loop Output-Feedback Monte Carlo Benchmark Under ±20% 7-Rate Mismatch, ±8% Initial Uncertainty, AR(1) Noise, and 5% 6.5σ Spikes ($\kappa(O_7) = 745.6$); **(B)** C99 Multi-Step-Size Quantization Deadband Ablation vs. IEEE-754 `FP64`.

A. Observer ($N_{MC} = 120$)	**Med. J [IQR]**	**$\hat{h}$ RMSE**	**Prem. Sw.**	$h(T_{max})$	$W(T_{max})$	**B. Step N (Δt)**	`FP32`	**Naive `Q16`**	`Q8.24`	`Q8.24+ΣΔ`
Oracle State (0.080)	0.909 [0.84, 0.98]	0.00%	0.00%	0.0255	1.0433	500 (0.180 d)	0.0001%	13.51%	0.602%	**0.0543**%
Standard EKF (0.080)	0.906 [0.85, 0.98]	2.18%	24.17%	0.0274	1.0411	1,000 (0.090 d)	0.0002%	23.31%	1.827%	**0.0563**%
Huber-EKF (0.080)	0.913 [0.86, 0.99]	1.95%	14.17%	0.0255	1.0435	2,000 (0.045 d)	0.0005%	39.61%	2.841%	**0.0554**%
Huber-EKF (0.065)	**0.933** [0.87, 1.00]	**1.93**%	**5.00**%	**0.0218**	**1.0479**	4,000 (0.0225 d)	0.0010%	84.41%	22.02%	**0.0534**%
Stage-1 $\sigma(O_7)$ = [963.4, 253.3, 205.7, 171.2, 137.4, 15.6, 1.29] ($\sigma_8 = 0$)						8,000 (0.0113 d)	0.0022%	100.0%	17.44%	**0.0539**%

Table 5: 1,000-Step 8-State `Q16.16` `RK4` Parity Against IEEE-754 `FP64` (`coralnpu_hil_testbench.c`) and Saltelli Global Sobol Sensitivity Indices on $\log_{10}(1 + J(\mathbf{u}))$ (2,560 ODE Runs).

State	**Term. `FP64`**	**`Q16.16` Int**	**`Q16.16` Float**	**Term. Err**	**Max / RMS Err**	**Sobol Param.**	**Range**	S_i	S_{T_i}
x_0: $W(T_{max})$	1.048706	68,729	1.048721	0.0015%	0.0028%/0.0015%	$k_{mito,D,max}$	[0.085, 0.145]	0.739	0.859
x_1: $D(T_{max})$	0.026781	1,755	0.026779	0.0074%	0.0464%/0.0137%	k_{clr}	[0.120, 0.240]	0.125	0.125
x_2: $B(T_{max})$	0.040012	2,623	0.040024	0.0296%	**0.0563**%/0.0301%	$k_{rep,adv}$	[1.050, 1.500]	0.102	0.138
x_3: $E(T_{max})$	0.630235	41,303	0.630234	0.0001%	0.0026%/0.0011%	θ_{GSH}	[1.020, 1.250]*	0.023	0.023
x_4: $G(T_{max})$	0.318642	20,883	0.318649	0.0022%	0.0047%/0.0020%	—	—	—	—
x_5: $N(T_{max})$	1.469839	96,326	1.469818	0.0014%	0.0063%/0.0023%	*Sub-baseline $\theta_{GSH} \in [0.92, 1.02]$ crosses			
x_6: $C(T_{max})$	1.299926	85,192	1.299927	0.0001%	0.0061%/0.0030%	ROS cliff ($S_{T_i} = 0.989$). C99 EKF: 0.94% RMSE.			
x_7: $M_8(T_{max})$	1.736594	113,816	1.736694	0.0058%	0.0154%/0.0085%				

reversal ($u_{2,target} \leftarrow 0, u_{1,target} \leftarrow 1$) caps $EP_{peak} = 1.521$, bounds $\Delta h_{max} = +0.0098$ (inside the analytical excursion bound 0.033 of Eq. A.16), and restores $W(T_{max}) = 1.0128$.

5.3 Host-Emulated C99 Fixed-Point Parity, Sobol Indices, and Memory/Power Budget

Executing `build/coralnpu/coralnpu_hil_testbench.c` (`-std=c99 -O2 -Wall -Wextra -Werror`; Tables 4B–6) confirms 0.0563% max 1,000-step `RK4` error across all 8 states, 0.94% $\hat{h}$ RMSE in closed-loop C99 Huber-EKF, and flat 0.0534%–0.0563% error across $N \in \{500, \ldots, 8{,}000\}$ (vs. 100% naive `Q16.16` deadband stall). Global Saltelli Sobol indices (2,560 ODE runs, Table 5) confirm that Stage-1 mitophagy capacity $k_{mito,D,max}$ dominates outcome variance ($S_i = 0.739, S_{T_i} = 0.859$) inside the physiological redox box, while Table 6 verifies 3,696 B Bank 0 residency and a 3,006-cycle (6.012 μs, 3.090 mW active burst at 500 MHz) compute profile.

6 Discussion, Limitations, and Reproducibility Manifest

Limitations and Scope of Claims. All ODE trajectories, discrete-adjoint optimal control schedules, mismatched EKF benchmarks, and C99 fixed-point results in this study are strictly computational and host-emulated *in silico* evaluations; no prospective wet-lab, animal, or clinical claims are made. Biological controllability and indirect observability exhibit hard physical limits: initial wild-type depletion inside Zone 0 (18.9% of the (W_0, k_{dmg}) grid) is irreversibly unrescuable without exogenous organelle transfer, and 4-channel proxy sensing leaves $\kappa(O_7) = 745.6$ pre-switch ill-conditioning requiring conservative threshold margin tightening ($\hat{h}_{gate} = 0.065$). Stochastic copy-number drift at very low mtDNA counts, spatial axonal reaction-diffusion PDEs, and physical RTL silicon synthesis lie outside the deterministic single-compartment scope of Paper I.

Table 6: Compiler-Verified Single-Bank L1 DTCM Memory Map (3,696 B in `Bank 0`) and Host-Emulated Cycle-Weighted Active NPU Power vs. External Sensor Rail (500 MHz, 22 nm ULP CMOS Model).

L1 DTCM Bank 0 Struct Group	**Bytes**	**% of 8 KB**	**Active NPU Compute Phase**	**Cycles**	**Frac.**	**Power / Weighted**
INT8 Gate (1,216) + Goertzel LUTs (768)	1,984 B	24.22%	1. RVV `Zve32x RK4` + Huber-EKF	1,840	61.21%	3.45 → 2.112 mW
Mixed `Q16/Q24` 7-State EKF ($\mathbf{P}, \mathbf{F}, \mathbf{Q}, \hat{\mathbf{x}}$)	452 B	5.52%	2. 16-Bin Goertzel & 64-Frame Gate	640	21.29%	2.60 → 0.554 mW
`Q8.24 RK4` (128) + $\Sigma\Delta$ (64) + Params (132)	324 B	3.96%	3. Supervisory Threshold & Rollback	526	17.50%	2.42 → 0.424 mW
64-Frame Ring (136) + $\mathbf{x}_{\text{ckpt}}$ (32) + Stack (768)	936 B	11.43%	**Total Compute Burst** (6.012 μs)	**3,006**	**100%**	**Mean:** 3.090 **mW**
Verified `sizeof` Active Bank 0	**3,696** B	**45.12%**	Deep-Sleep Retention (Banks 1–3 Gated)	—	> 99.99%	18.0 μW cont.
Unallocated Bank 0 Headroom	4,496 B	54.88%	External Sensor/ADC (15 ms/frame)	—	Sep. rail	1.800 mW window

MIASE Reproducibility Manifest. Following MIASE guidelines [21], every figure, table entry, and numerical metric is deterministically regenerated from source via `python3 simulator/generate_paper1_verified_benchmarks.py` (seed 20261109) and the freestanding C99 testbench `build/coralnpu/coralnpu_hil_testbench.c` (repository: https://github.com/deesatzed/mitorestorer).

A Complete 8-State Equations and Mathematical Proofs

A.1 Complete 8-State System, Auxiliary Rates, and Observation Map

To guarantee exact 1-to-1 parity between LaTeX, `eval_rhs_fp64` in Python, and `coralnpu_eval_core8_rhs_q16` in C99, the closed-form auxiliary bioenergetic rates are:

$$V_{\text{lys}}(B) = \frac{K_B}{K_B + B}, \qquad k_{\text{bio}}(u_2, N) = k_{\text{bio},0} + (k_{\text{bio,max}} - k_{\text{bio},0})\, u_2\, \frac{N}{1+N}, \tag{A.1}$$

$$k_{\text{mito},D}(u_1) = k_{\text{mito},W} + (k_{\text{mito},D,\text{max}} - k_{\text{mito},W})\, u_1, \qquad \text{ATP}(\mathbf{x}) = (W + \eta_D D)(1 + \gamma_C C), \tag{A.2}$$

$$\dot{e}_R(\mathbf{x}) = \frac{1 + \beta_B B + \beta_h h}{\max(0.10,\ W + \eta_D D)(1 + \gamma_C C)}, \qquad f_{\text{eff}}(u_1, u_2) = 1 - c_1 u_1 + c_2 u_2, \tag{A.3}$$

$$\text{EP}(\mathbf{x}, \mathbf{u}) = \dot{e}_R(\mathbf{x})\, f_{\text{eff}}(u_1, u_2)^2, \qquad k_{\text{bio},0} = \frac{k_{\text{mito},W}}{1 - 1/K}. \tag{A.4}$$

Together with Eqs. (1)–(3) for (W, D, B), the remaining five state equations for filtered energy potential E, secreted stress marker G, redox cofactor N, phosphagen buffer C, and epigenetic drift readout M_8 are:

$$\dot{E} = k_E\big(\text{EP}(\mathbf{x}, \mathbf{u}) - E\big), \qquad \dot{G} = k_{G,\text{prod}}\, E - k_{G,\text{clr}}\, G, \tag{A.5}$$

$$\dot{N} = k_{N,\text{syn},0} + k_{N,\text{syn},2}\, u_2 - k_{\text{PARP}} \max(0, E - \theta_{\text{GSH}}) N - k_{N,\text{use}}\, N, \tag{A.6}$$

$$\dot{C} = (k_{\text{oct},0} + k_{\text{oct},2}\, u_2)(1.30 - C) - k_{\text{ox}} \max(0, E - \theta_{\text{cys}}) C, \tag{A.7}$$

$$\dot{M}_8 = k_{\text{drift}}\, E\, (10.0 - M_8) - k_{\text{TET}}\, u_2\, N\, C\, M_8. \tag{A.8}$$

At discrete controller steps t_k, the 4-channel proxy observation vector and post-switch redox safety condition ($u_{1,\text{S2}} = 0.35$) are:

$$\mathbf{y}_k = \big[E(t_k),\ \text{ATP}(\mathbf{x}(t_k)),\ G(t_k),\ N(t_k)\big]^\top + \mathbf{v}_k, \tag{A.9}$$

$$\text{EP}(t^+_{\text{gate}}) = \frac{1 + \beta_B B(t_{\text{gate}}) + \beta_h h_{\text{gate}}}{W(t_{\text{gate}})\big[1 + \eta_D h_{\text{gate}}/(1 - h_{\text{gate}})\big]\big[1 + \gamma_C C(t_{\text{gate}})\big]}\, (1 - c_1 u_{1,\text{S2}} + c_2)^2 \le \theta_{\text{GSH}}. \tag{A.10}$$

A.2 Quotient-Rule Derivation and Proofs of Lemma 1 and Proposition 1

Applying the quotient rule $\dot{h} = (\dot{D}W - D\dot{W})/M^2$ to Eqs. (1)–(2), let $d_{\text{ros}} \triangleq k_{\text{dmg}} \max(0, \text{EP} - \theta_{\text{GSH}})$. The $+d_{\text{ros}}W$ source term in $\dot{D}$ contributes $(W^2/M^2)d_{\text{ros}} = (1-h)^2 d_{\text{ros}}$, while the $-d_{\text{ros}}W$ loss term in $\dot{W}$ contributes $+(DW/M^2)d_{\text{ros}} = h(1-h)d_{\text{ros}}$; their sum is $(1-h)d_{\text{ros}}$, yielding Eqs. (4)–(5).

Define the boundary and interior critical bottleneck thresholds:

$$B^*_{\text{crit}} \triangleq K_B\left(\frac{k_{\text{mito},D,\max}}{k_{\text{rep,adv}}\, k_{\text{bio},0}\,(1 - W^*/K)} - 1\right), \qquad B_{\text{crit,mono}} \triangleq K_B\left(\frac{k_{\text{mito},D,\max}}{k_{\text{rep,adv}}\, k_{\text{bio},0}\,\big[1 - (W^* - D_0)/K\big]} - 1\right). \tag{A.11}$$

Proof of Lemma 1. Under $(u_0, u_1, u_2) = (1, 1, 0)$ and $\sup_{\Omega_1} \text{EP} \le \min(\theta_{\text{GSH}}, E_{\max})$, $d_{\text{ros}} \equiv 0$ on Ω_1. Checking Nagumo's subtangent condition on each face of Ω_1: $\dot{E}|_{E=0} \ge 0$, $\dot{E}|_{E=E_{\max}} \le 0$, $\dot{C}|_{C=1.30} \le 0$, $\dot{C}|_{C=C_{\min}} \ge 0$ (when $k_{\text{oct},0}(1.30 - C_{\min}) \ge k_{\text{ox}} \max(0, E_{\max} - \theta_{\text{cys}})C_{\min}$), $\dot{B}|_{B=B_{\text{eq}}} = 0$, and $\dot{B}|_{B=B_0} = k_{\text{clr}}(B_{\text{eq}} - B_0) \le 0$. On the upper mutant face $D = D_0$, $W \ge W^* - D_0 \implies M = W + D_0 \ge W^*$, so $\dot{D}|_{D=D_0} \le D_0[k_{\text{rep,adv}} k_{\text{bio},0}(1 - W^*/K) - k_{\text{mito},D,\max} K_B/(K_B + B)] \le 0$ for all $B \le B^*_{\text{crit}}$. On the lower wild-type face $W = W_{\min} = W^* - D_0$, $D \le D_0 \implies M \le W^*$, so $\dot{W}|_{W_{\min}} \ge W_{\min}[k_{\text{bio},0}(1 - W^*/K) - k_{\text{mito},W}] = 0$ by Eq. (A.4), while $\dot{W}|_{W=K} < 0$. Hence Ω_1 is jointly positively invariant. Integrating $\dot{B} = b_{\text{basal}} - k_{\text{clr}}B$ yields t_0 in Case 2. On the canonical neuronal box ($W \in [0.60, 1.25], D \in [0, 0.40], B \in [0.04, 0.50], C \in [0.65, 1.30], E_{\max} = 1.35$), $\partial\text{EP}/\partial D > 0$ on $[0, 0.40]$, so EP attains its exact maximum at $(0.60, 0.40, 0.50, 0.65)$: $\max_{\Omega_1} \text{EP} = [1.385/(0.660 \times 1.221)] \times 0.6084 = 1.0456 < 1.150$, with $B^*_{\text{crit}} = 1.2020 > 0.500$ and $B_{\text{crit,mono}} = 0.3085$ (crossed at $t = 3.16$ d). □

Proof of Proposition 1. On $\Omega_1(D_0, B_0)$, $d_{\text{ros}} \equiv 0$, $B(t) \le B_0$, and $M(t) \ge W^* - D_0$, so Eq. (5) satisfies $\lambda_h(t) \ge \lambda_{\min}$ with

$$\lambda_{\min} \triangleq \frac{k_{\text{mito},D,\max} K_B}{K_B + B_0} - k_{\text{mito},W} - (k_{\text{rep,adv}} - 1)k_{\text{bio},0}\left(1 - \frac{W^* - D_0}{K}\right). \tag{A.12}$$

Separating variables in $\frac{d}{dt}\ln[h/(1-h)] = -\lambda_h(t) \le -\lambda_{\min}$ and noting $1 - h(t) \ge 1 - h_0$ yields Eq. (6). Linearizing $\dot{D}$ at $(W^*, 0, B_{\text{eq}})$ where $k_{\text{bio}}(1 - W^*/K) = k_{\text{mito},W}$ gives $\dot{D} \approx \mu_D D$. □

A.3 Closed-Form Bottleneck Integral and Proof of Proposition 2

Proof of Proposition 2. Integrating $\frac{d}{dt}\ln[h/(1-h)] = -\lambda_h(t)$ over $[0, t]$ gives $\ln[h(t)/(1-h(t))] = \ln[h_0/(1-h_0)] - s(t)$, which inverts to Eq. (8). During Stage 1 ($u_0 = 1, u_1 = 1, u_2 = 0$), $k_{\text{bio}}(0, N) \equiv k_{\text{bio},0}$ and $B(q) = B_{\text{eq}} + (B_0 - B_{\text{eq}})e^{-k_{\text{clr}}q}$. Let $T_{1,\text{apriori}} \triangleq s^*(h_0, h_{\text{gate}})/[k_{\text{mito},D,\max} V_{\text{lys}}(B_{\text{eq}}) - k_{\text{mito},W} - (k_{\text{rep,adv}} - 1)k_{\text{bio},0}(1 - M_0/K)]$. Integrating $V_{\text{lys}}(B(q))$ via $w = e^{k_{\text{clr}}q}$ gives the closed-form *a priori* mean competence and constant rescaling rate:

$$\bar{V}_{\text{lys}} \triangleq \frac{1}{T_{1,\text{apriori}}}\int_0^{T_{1,\text{apriori}}} \frac{K_B\, dq}{K_B + B(q)} = \frac{K_B}{K_B + B_{\text{eq}}}\left[1 - \frac{\ln\big[(K_B + B_0)/\big(K_B + B(T_{1,\text{apriori}})\big)\big]}{k_{\text{clr}}\, T_{1,\text{apriori}}}\right], \tag{A.13}$$

$$\hat{\lambda}_0 \triangleq k_{\text{mito},D,\max}\, \bar{V}_{\text{lys}} - k_{\text{mito},W} - (k_{\text{rep,adv}} - 1)k_{\text{bio},0}\left(1 - \frac{M_0}{K}\right). \tag{A.14}$$

Subtracting Eq. (A.14) from Eq. (5) yields the integrated logit error bound

$$\big|s(t) - \hat{\lambda}_0 t\big| \le \underbrace{k_{\text{mito},D,\max}\int_0^t \big|V_{\text{lys}}(B(q)) - \bar{V}_{\text{lys}}\big|\, dq}_{\mathcal{E}_B(t)} + \underbrace{(k_{\text{rep,adv}} - 1)\frac{k_{\text{bio},0}}{K}\int_0^t \big|M(q) - M_0\big|\, dq}_{\mathcal{E}_M(t)}, \tag{A.15}$$

and since $|d(e^{\ell}/[1+e^{\ell}])/d\ell| = h(1-h) \le 1/4$, Eq. (9) follows immediately. □

A.4 Proof of Proposition 3 and Actuator-Reversal Excursion Bound

Proof of Proposition 3. Any 64-frame window overlapping an L_{dist}-frame burst contains at most $c_0 + L_{\text{dist}}$ ones, which is strictly below 60 whenever $L_{\text{dist}} \le 59 - c_0$. For stationary noise, the latent Gaussian process e_k is a reversible Markov chain on $(\mathbb{R}, \mathcal{N}(0,1))$ with L^2 spectral gap $\gamma_{\text{spec}} = 1 - \rho$. Applying Theorem 3.3 of Paulin [20] to the bounded indicator $f(e_k) = I_k \in \{0,1\}$, multiplying by the exact conditioning factor $p_{z|Q}$, and applying Boole's union bound over K_{frames} steps yields Eq. (10). □

Finally, during an acute post-switch insult over $[t_{\text{insult}}, t_{\text{insult}}+\Delta t_{\text{active}}]$ where $\Delta t_{\text{active}} \approx \delta t_{\text{detect}} + \tau_{\text{wash}} \ln[(\text{EP}_{\text{peak}} - \theta_{\text{GSH}})/(\theta_{\text{abort}} - \theta_{\text{GSH}})]$, integrating the positive ROS term in Eq. (4) from $h(t_{\text{insult}}) \le h_{\text{gate}}$ bounds the net heteroplasmy excursion prior to actuator washout by:

$$\Delta h_{\max} \le (1 - h_{\text{gate}})\, k_{\text{dmg}} \int_{t_{\text{insult}}}^{t_{\text{insult}}+\Delta t_{\text{active}}} \max\bigl(0,\, \text{EP}(q) - \theta_{\text{GSH}}\bigr)\, dq. \tag{A.16}$$

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